
Empirical Characterization of Kubernetes HPA Limitations for Stateful WebSocket Workloads

Abhash Behera · Aditya Hota · Suprit Kumar Naik · Ritesh Kumar Singh

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Abstract The Kubernetes Horizontal Pod Autoscaler (HPA) implements a proportional control law over CPU utilization as its primary scaling signal. An architecture that performs reliably for stateless, request-driven workloads but is poorly matched to stateful, persistent-connection protocols such as WebSocket. In WebSocket deployments, the primary unit of computational commitment is an active network connection whose session state persists independently of CPU activity; the termination of any pod severs the connections it hosts. This paper presents a systematic empirical characterization of HPA failure under WebSocket workloads through a progressive chain of four controlled experiments. Experiment A establishes a correct-behavior baseline under the idealized condition of CPU-to-connection proportionality. Experiment B1 demonstrates that HPA over-provisions by 87.5% under cyclic load, exhausting `maxReplicas` within 99 seconds and requiring 650 seconds to recover. Experiment B2-Instrumented uses Prometheus instrumentation to measure reconnection storm rates causally linked to HPA scale-down decisions, with a peak observed rate of 1,400 connections/second and a 51.9%

connection count overshoot (1,215 vs. 800 target). Experiment B3 demonstrates permanent, irrecoverable staircase connection destruction under non-reconnecting clients, with the active session population monotonically reduced from 800 to a mean of 649 (± 8.8) across five replications, representing 18.8% mean connection loss with zero recovery. Our experiments suggest that these failure modes are *architectural* rather than configurational: they arise from the limitations of CPU as a scaling signal for persistent-connection workloads and are unlikely to be remediated by HPA parameter tuning alone. The findings motivate the design of connection-aware autoscaling policies, as pursued in the companion paper.

Keywords Kubernetes · Horizontal Pod Autoscaler · WebSocket · Stateful Workloads · Autoscaling · Reconnection Storm · Cloud Computing · Performance Evaluations

1 Introduction

Kubernetes has established itself as the de facto standard platform for container orchestration, providing declarative workload management, automated scheduling, and elastic scaling for cloud-native applications [1, 12]. Central to its elasticity model is the **Horizontal Pod Autoscaler (HPA)**, which evaluates observed resource metrics at a fixed synchronization interval and adjusts replica counts through a proportional control law [5, 9]. For stateless, request-driven workloads such as HTTP APIs, where instantaneous CPU consumption is a reliable proxy for current demand, this approach converges stably and performs predictably [2, 8].

A fundamental shift in cloud-native workload characteristics, however, increasingly invalidates this assumption. Real-time communication systems, IoT telemetry pipelines, collaborative editing platforms, financial

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data streams, and multiplayer gaming backends rely on the WebSocket protocol [4] - a full-duplex, persistent-connection transport layer in which each client maintains a long-lived TCP session with the server. In such workloads, the primary unit of computational commitment is not a CPU cycle or a transient HTTP transaction, but an *active network connection* whose session state may persist for minutes to hours. The termination of any pod severs the connections it hosts, triggering a forced client reconnection cascade - a phenomenon we term a **reconnection storm**.

This creates a significant mismatch: **CPU utilization does not encode connection state fidelity by default**. A server sustaining 800 idle but active WebSocket sessions may exhibit near-zero CPU utilization, yet the destruction of a single pod hosting those sessions is disruptive to the affected clients. HPA does not distinguish an *idle-but-connected* server from a *genuinely unloaded* one, and will therefore terminate pods that are essential to live client sessions.

Prior investigations of WebSocket autoscaling, such as Dickel et al. [3], have acknowledged the additional complexity of stateful protocols compared to stateless HTTP workloads. However, that work neither quantifies the failure modes of CPU-based HPA with empirical precision, nor provides a measured reconnection storm rate. This paper addresses the characterization gap through the following contributions:

1. **Progressive four-experiment evidence chain.** Experiments A, B1, B2-Instrumented, and B3 form a controlled sequence that isolates and quantifies each HPA failure mode, progressing from over-provisioning, to reconnection storm measurement, to permanent connection destruction.
2. **Empirical measurement of reconnection storm magnitude.** We observe a peak reconnection storm rate of 1,400 connections/second as a direct causal consequence of an HPA scale-down decision, using Prometheus instrumentation to provide a precise, reproducible measurement.
3. **Causal evidence of architectural limitation.** Experiment B3 demonstrates the causal chain: CPU drops to near-zero (connections become idle) → HPA scales down → active sessions are permanently destroyed. This is a direct consequence of the CPU signal's inability to encode connection state.
4. **Establishment of the stabilization window semantic mismatch.** We show that HPA's built-in stabilization window, which buffers metric-derived replica recommendations, is insufficient to protect idle connections: with persistent near-zero CPU, every window entry recommends minReplicas, and the stabilized output converges to minReplicas regardless of window duration.

5. Quantified failure metrics as comparison baselines.

All failure modes are quantified at connection-second granularity (87.5% over-provisioning, 1,400 conn/s peak storm, 18.8% mean staircase connection loss), providing reproducible baselines for future autoscaling research in this domain.

The remainder of this paper is structured as follows. Section 2 surveys related work. Section 3 provides a concise overview of the HPA control architecture. Section 4 presents the four-experiment evidence chain. Section 5 analyzes the root causes and implications. Section 6 discusses limitations. Section 7 concludes and motivates future work.

2 Related Work

2.1 Kubernetes HPA Architecture and Behavior

Nguyen et al. [9] provide a comprehensive architectural analysis of the Kubernetes HPA feedback loop, characterizing the behavior of both Kubernetes Resource Metrics (KRM) and Prometheus Custom Metrics (PCM) pipelines across diverse scaling configurations. Their work demonstrates that metric pipeline choice - particularly the Prometheus scrape interval - materially affects HPA responsiveness and scaling accuracy. The present paper extends their analysis to the *stateful workload* domain, a scenario their framework did not evaluate.

Casalichio and Perciballi [2] demonstrated that the choice between relative and absolute metrics directly determines scaling stability, but confined their evaluation to CPU and memory signals for HTTP workloads. Llorido-Botrán et al. [8] provide a broader survey of auto-scaling techniques in cloud environments, establishing the theoretical framing of proportional controllers that this paper applies to persistent-connection workloads.

2.2 Custom and Application-Level Metrics for HPA

Kubernetes supports application-defined scaling signals through the Prometheus Adapter [7], enabling HPA to react to metrics beyond CPU. Qu et al. [10] surveyed HTTP-rate-based autoscaling; Rossi et al. [11] studied hybrid CPU-and-traffic strategies. However, all these formulations presuppose that the scaling metric is *stateless across pod boundaries* - an assumption violated by connection-holding protocols where session state is affined to a specific pod instance. Furthermore, even when HPA is configured with a correct connection-count metric via the Prometheus Adapter, HPA's stabilization semantics remain CPU-oriented: the window stabilizes metric-derived replica recommendations rather than connection-derived desired

counts. This distinction is analyzed in Section 5 and evaluated experimentally in the companion paper.

2.3 Stateful Workloads and Protocol-Aware Autoscaling

Dickel et al. [3] identified stateful autoscaling as an open research challenge specific to IoT gateway deployments using WebSocket and MQTT. While their work motivates the need for protocol-aware scaling and proposes monitoring active connections via Prometheus, it provides no quantitative characterization of HPA failure modes and no controller-level solution. This paper fills the characterization gap; the companion paper fills the solution gap.

2.4 Graceful Connection Draining

Production Kubernetes deployments commonly use `preStop` hooks and `terminationGracePeriodSeconds` to allow stateless pods to drain in-flight requests before receiving SIGKILL. For WebSocket workloads with session durations of minutes to hours, this mechanism provides insufficient time for graceful drain. The 30-second limbo window between SIGTERM and SIGKILL is explicitly characterized in Experiment B3 (Section 4.4) as a design constraint, not a tunable parameter.

2.5 Research Gap

To the best of our knowledge, no prior work has simultaneously: (i) empirically quantified the specific failure modes of CPU-based HPA for persistent WebSocket connections at the connection-second granularity; (ii) measured reconnection storm magnitude as a causal consequence of HPA scale-down decisions; and (iii) established through controlled experiments that the failure is architectural rather than configurational. This paper addresses all three dimensions.

3 Background: HPA Control Architecture

This section provides a concise overview of the Kubernetes HPA control loop and its underlying metric pipelines. For a detailed characterization of HPA behavior under stateless HTTP workloads using both Kubernetes Resource Metrics (KRM) and Prometheus Custom Metrics (PCM) pipelines, we refer the reader to Nguyen et al. [9]. Our preliminary baseline experiments with both pipelines replicated their key findings: staircase scaling dynamics with actuation delay during scale-up, CPU overshoot at burst onset, and diminishing returns from reducing the Prometheus scrape

interval below 30 seconds. These baselines confirmed that HPA converges stably for stateless workloads and informed the experimental design of the subsequent failure characterization.

Figure 1 illustrates the HPA feedback loop. The controller implements the following discrete-time proportional control law evaluated every 15 seconds (the default synchronization period):

$$desiredReplicas = \left\lceil currentReplicas \times \frac{currentMetricValue}{targetMetricValue} \right\rceil \quad (1)$$

This formulation implicitly assumes that `currentMetricValue` faithfully reflects workload intensity across all running replicas. For CPU-based scaling of stateless workloads, this assumption generally holds. For persistent-connection workloads, it breaks down: idle connections consume negligible CPU yet represent committed session state whose disruption is visible to end clients. The experimental question is therefore: **what are the consequences of this control design when the assumed signal-to-demand coupling is absent?**

4 Empirical Evidence of HPA Failure for Stateful WebSocket Workloads

We present a controlled, progressive chain of experiments designed to isolate and characterize each failure mode of CPU-based HPA when applied to persistent WebSocket workloads. The experiments are ordered from least severe (over-provisioning) to most severe (permanent connection destruction), with each experiment building upon the instrumentation and findings of its predecessor.

All WebSocket experiments share a common infrastructure: a 3-node Kind cluster (1 control-plane + 2 worker nodes, Kubernetes v1.31.6, kind v0.25.0, configured via `kind.yml`) [6]. The Metrics Server is patched with `--metric-resolution=15s` and `--kubelet-insecure-tls` (required for kind’s self-signed kubelet certificates; not appropriate for production). All experimental scripts, cluster configuration, server code, load generator, and raw results are available at: <https://github.com/MistaHolmes/ws-k8-controller.git>. The shared experimental infrastructure comprises: a custom Python `asyncio` WebSocket server (image: `ws-instrumented`), 800 persistent client connections established by the load generator, and Kubernetes HPA configured with `targetCPUUtilizationPercentage: 60` unless stated otherwise.

Instrumented server: definition and rationale By “instrumented” we mean the server was extended to expose

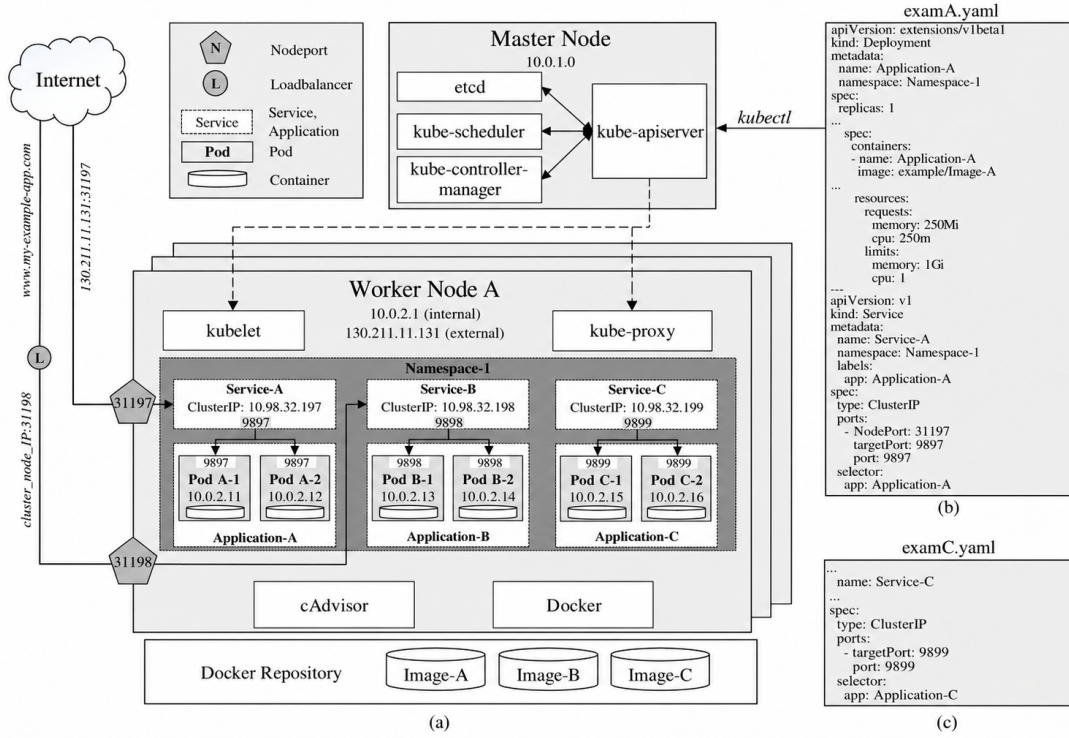


Fig. 1 Standard Kubernetes HPA feedback loop architecture, reproduced from Nguyen et al. [9]. The HPA controller periodically reads aggregated resource metrics from the API server (sourced either from the Metrics Server for KRM or the Prometheus Adapter for PCM) and computes a desired replica count using Equation 1. This loop assumes the observed metric faithfully represents current workload intensity - an assumption that breaks down for persistent-connection workloads where connections persist independently of CPU utilization.

Prometheus-format metrics (notably a gauge `active_connections` and a counter `new_connections_total`) and a minimal control endpoint (for example, `/drain`) to enable graceful handoff. This instrumentation was adopted to obtain direct, low-latency visibility into live session counts and connection churn - signals that CPU does not capture. The approach (i) supplies a precise measurement signal for storm rate computation, (ii) permits reproducible measurement of reconnection storms via PromQL `rate()` on `new_connections_total`, and (iii) is minimally invasive and portable.

4.1 Experiment A: HPA Baseline Under Monotonic Correlated Load

Hypothesis. Under steady, monotonically increasing WebSocket load in which each connection actively generates CPU work ($\text{CPU_WORK}=1$), HPA will correctly scale the deployment, establishing an empirical baseline for comparison with subsequent experiments.

Experimental Setup. Eight hundred WebSocket clients connect incrementally over 300 seconds. Each server-side message receipt triggers a bounded CPU spin loop, maintaining strict proportionality between active connection

count and CPU utilization: $\text{CPU} \propto \text{connections}$. HPA was configured with `minReplicas=2`, `maxReplicas=10`, and `targetCPUUtilizationPercentage=60`.

Results. As shown in Figure 2, HPA scaled the deployment monotonically from 2 to 5 replicas as CPU rose proportionally with connection count. Following complete load removal, the system enforced the default 300-second scale-down stabilization window before descending cleanly to `minReplicas`. No oscillatory behavior or reconnection disruption was observed.

The replica timeline (Figure 2) confirms that HPA operates correctly *when its signal accurately encodes workload demand*. Crucially, this experiment represents a best-case scenario that is rarely achievable in production WebSocket deployments, where connection activity levels are typically decoupled from CPU consumption.

Conclusion. CPU-based HPA functions correctly only under the idealized condition of perfect CPU-to-connection proportionality. Subsequent experiments systematically invalidate this condition.

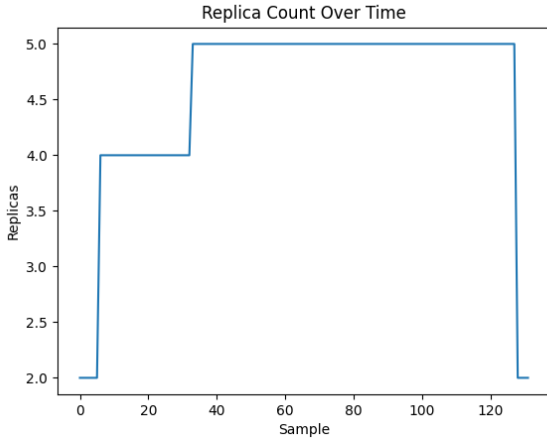


Fig. 2 Experiment A: Replica count timeline under monotonic WebSocket load with $\text{CPU} \propto \text{connections}$. The clean staircase ascent (2→5), flat stabilization plateau, and controlled descent to the minimum represent the theoretical ideal for HPA - achievable only under perfect signal-to-demand correlation.

4.2 Experiment B1: HPA Under Cyclic Load - Over-Provisioning and Recovery Asymmetry

Hypothesis. Under realistic cyclic HIGH/LOW load patterns that simulate natural connection activity fluctuations, HPA will exhibit systematic over-provisioning and asymmetrically slow resource recovery.

Experimental Setup. Load alternates between HIGH phases (800 connections, active CPU-generating pinging, 60-second duration) and LOW phases (connections held persistently idle, no pinging activity, 30-second duration). Crucially, during LOW phases the 800 client connections remain open; only the periodic ping messages cease, collapsing CPU utilization toward zero while connection state is preserved. Parameters were adjusted to $\text{maxReplicas}=15$ to allow unconstrained scale-up. The HPA stabilization window was kept at the Kubernetes default of 300 seconds for scale-down ($\text{stabilizationWindowSeconds}: 300$), intentionally unmodified to evaluate HPA behavior under factory settings for cyclic load. Five consecutive HIGH/LOW cycles (60 s HIGH + 30 s LOW = 90-second period) were run, for a total active cycle phase of 450 seconds.

Results. During HIGH phases, HPA exhausted $\text{maxReplicas}=15$ within 99 seconds - a factor of $1.875 \times$ the theoretically correct value of 8 replicas ($\lceil 800/100 \rceil$). The core mechanism driving over-provisioning is the interaction between the 90-second cycle period and the 300-second stabilization window: each LOW phase (30 s) ends before the window has elapsed, so every new HIGH phase begins with the replica count still at or near the ceiling from the previous cycle. Over five cycles, the replica count

ratchets to maxReplicas and remains there, with each brief LOW phase failing to provide enough sustained low-CPU time for HPA to authorize a descent. Convergence to minReplicas following the final cycle required approximately 650 seconds. The replica count timeline (Figure 3) exhibits the expected ratcheting saturation pattern.

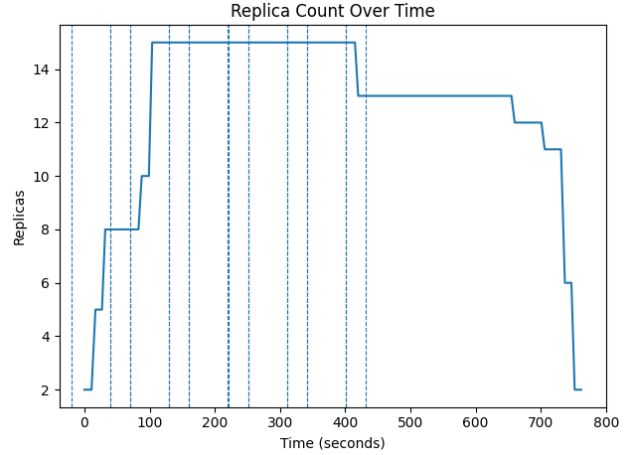


Fig. 3 Experiment B1: HPA saturates at $\text{maxReplicas}=15$ within 99 seconds under cyclic load, representing an 87.5% over-provision relative to the connection-derived optimum of 8 replicas. Recovery to the minimum requires ~650 seconds, demonstrating severe asymmetry between scale-up and scale-down responsiveness.

Conclusion. Cyclic workloads expose HPA’s inability to distinguish connection-holding periods (where CPU is low but capacity is required) from genuinely idle periods (where capacity can be safely released). The 87.5% over-provisioning rate and disproportionate recovery time (~650 s) represent significant resource efficiency penalties in production deployments.

4.3 Experiment B2-Instrumented: Quantifying Reconnection Storm Intensity

Methodological note: supersession of a non-instrumented pilot Prior to this experiment, a preliminary non-instrumented variant (B2-Extended LOW) was conducted in which the LOW phase was extended beyond the default 300-second stabilization window to force HPA to $\text{minReplicas}=2$. While this pilot confirmed that active connections were being severed during scale-down events through qualitative observation of client disconnects, it produced no quantitative data: the server exposed no Prometheus metrics, connection loss counts were unverifiable, and reconnection rates were entirely unobservable. The pilot served its intended engineering purpose - confirming the failure mode was reproducible

before investing in full instrumentation - but it contributes no independent publishable evidence beyond what B2-Instrumented already provides with greater precision. Accordingly, B2-Extended LOW is not reported as a standalone experiment in this paper. Its sole scientific contribution is as the direct motivation for adding the Prometheus instrumentation described below.

Hypothesis. Each HPA scale-down event, by terminating pods hosting active connections, will trigger a measurable reconnection storm with peak rates exceeding 1,000 connections/second, and will induce overshoot of the steady-state connection target due to the cascade of concurrent reconnection attempts.

Experimental Setup. The WebSocket server was instrumented with two Prometheus metrics: `active_connections` (a gauge reflecting current session count) and `new_connections_total` (a monotonically increasing counter enabling rate computation via `rate()`). Prometheus was configured to scrape at a 15-second interval. The HPA policy was configured with `stabilizationWindowSeconds: 0` for scale-up and `stabilizationWindowSeconds: 60` for scale-down. Setting scale-up to zero ensures that each reconnection storm (which spikes CPU) immediately triggers scale-up, guaranteeing that every cycle produces a complete and observable scale-up/scale-down sequence within the measurement window. Each cycle consisted of a 60-second HIGH phase (800 pinging clients, `CPU_WORK=1`) followed by a 90-second LOW phase (connections idle, no pinging). The 90-second LOW duration was specifically chosen to exceed the 60-second scale-down stabilization window, ensuring that HPA always executes at least one scale-down event per cycle. Five complete HIGH/LOW cycles were executed.

Results. Every HPA scale-down event across all five cycles triggered a distinct, measurable reconnection storm. Storm rates were highly consistent: Cycle 1: 1,400.9 conn/s; Cycle 2: 1,298.3 conn/s; Cycle 3: 1,399.5 conn/s; Cycle 4: 1,251.8 conn/s; Cycle 5 trailed as the connection population had partially degraded. The peak reconnection rate reached **1,400 connections/second**, as shown in Figure 4. Concurrent with each storm, the active connection count transiently overshoot the 800-connection steady-state target, peaking at **1,215 connections** - a 51.9% overshoot (Figure 5).

The socket-level mechanism driving this overshoot is as follows: when a pod is sent `SIGKILL` after its 30-second termination grace period, its TCP connections are abruptly reset (TCP RST). All 800 clients detect closure simultaneously and immediately initiate reconnection. The Prometheus gauge `active_connections` on the surviving pods begins counting new incoming connections before the

OS completes cleanup of socket state (including the `TIME_WAIT` window for recently reset connections), transiently reporting a total above the 800-client ceiling. The overshoot resolves within 15–30 seconds as OS-level socket reclaim completes.

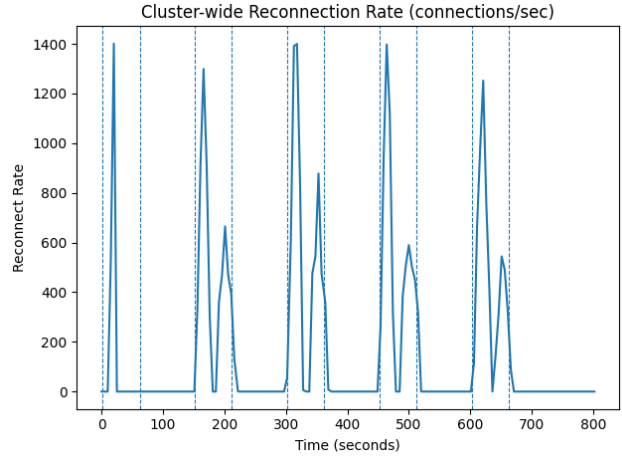


Fig. 4 Experiment B2-Instrumented: Prometheus-measured reconnection rate (`rate(new_connections_total[15s])`) over the five-cycle experiment. Each reconnection spike is precisely synchronized with an HPA scale-down event, with peak observed rates of 1,400 connections/second.

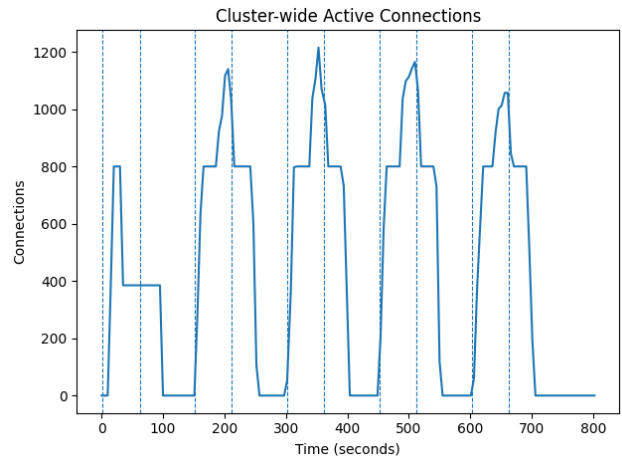


Fig. 5 Experiment B2-Instrumented: Active connection count during the reconnection storm period. The connection overshoot of 1,215 (51.9% above the 800-client target) arises as reconnecting clients establish new sessions before the server has fully released the state associated with severed connections.

Figure 6 provides the combined view of replica oscillation and connection disruption, establishing the causal chain from HPA scaling decision to client-side session disruption.

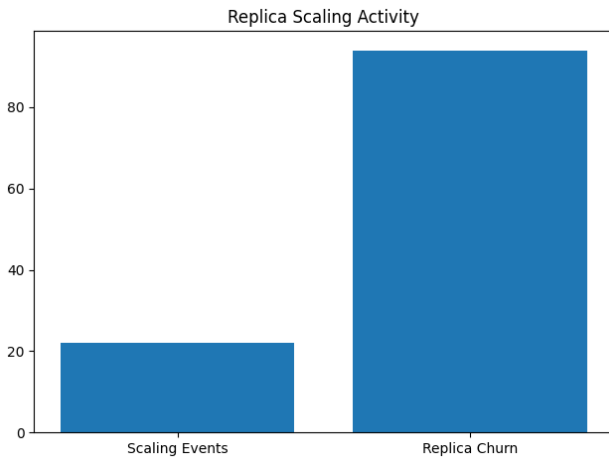


Fig. 6 Experiment B2-Instrumented: Temporal correlation between HPA replica changes (green) and active connection count (blue). Each downward step in replicas corresponds to a measurable spike in the connection disruption signal, confirming the direct causal relationship.

Conclusion. In our experiments, every HPA scale-down event was destructive to WebSocket client sessions. Each replica reduction severed all connections hosted on the terminated pod, producing a reconnection storm whose intensity scaled with the connection density of the removed replica. At 1,400 conn/s, such storms represent a significant degradation event for latency-sensitive applications.

4.4 Experiment B3: Permanent Connection Loss Under Non-Reconnecting Clients

Hypothesis. When WebSocket clients do not implement reconnection logic - modeling clients that treat connection loss as a terminal failure (e.g., batch data collectors, embedded IoT sensors, or long-running streaming sessions) - HPA scale-down events will cause *permanent, irrecoverable reduction* in the total session population.

Experimental Setup. The load generator was reconfigured with `reconnect: false`. Once a client's connection is severed by pod termination, the client neither retries nor reconnects. The HPA policy mirrored B2-Instrumented: `stabilizationWindowSeconds: 0` for scale-up and `stabilizationWindowSeconds: 60` for scale-down. CPU work was retained (`CPU_WORK=1`) to maintain HPA activity. The IDLE phase runs for up to 240 seconds (`IDLE_MAX_DURATION=240`), providing sufficient headroom for HPA to execute multiple sequential scale-down events.

Pod termination mechanics and the 30-second limbo
Understanding the exact Kubernetes pod termination

sequence is essential for interpreting this experiment. When HPA reduces `spec.replicas`, the following steps occur in order: (1) the target pod is immediately removed from the Service's endpoint slice, so no new connections are routed to it; (2) Kubernetes sends `SIGTERM` to the pod's main process and starts the `terminationGracePeriodSeconds` timer (default: 30 s); (3) during the 30-second grace period, existing TCP connections on the terminating pod remain open and functional - they are maintained by the OS kernel, not Kubernetes; (4) after `terminationGracePeriodSeconds` elapses, Kubernetes sends `SIGKILL`, which tears down the process and the OS forcibly resets all open TCP connections (TCP RST). This sequence is why the active connection drop visible in Figure 7 lags behind HPA's scale-down decision by approximately 30 seconds: the connection is not immediately destroyed when HPA acts, but is destroyed 30 seconds later when `SIGKILL` fires. This delay does not mitigate the harm - the connection still dies, the client still receives a TCP RST, and with `reconnect: false`, the session is permanently lost.

Results. The active connection count (Figure 7) exhibits a characteristic **staircase step-down** pattern. Each step corresponds precisely to an HPA scale-down decision: when a pod is terminated, the full contents of its connection table are permanently destroyed. Successive scale-down events cause monotonically fewer active connections, with no mechanism for recovery. The representative run (Run 4 of the five-run replication; see Table 1) exhibits the trajectory $800 \rightarrow 771 \rightarrow 708 \rightarrow 644$, a 19.5% permanent loss.

The combined view in Figure 8 illustrates the causal chain: (1) connections become idle and cease to generate CPU load; (2) HPA, observing near-zero CPU, decides to scale down; (3) the terminated pod destroys all connections it hosts permanently. This behavior is not addressable through configuration parameter changes - it arises from HPA's inability to observe connection state as a first-class scaling signal.

Conclusion. The reliance of HPA on CPU as a scaling signal causes the autoscaler to act against the interests of idle-but-connected stateful workloads. Our findings suggest that the problem is unlikely to be remediated by parameter tuning alone and motivates the exploration of alternative scaling architectures.

Replication and Statistical Validation To ensure the observed staircase destruction is robust and not a transient artifact, Experiment B3 was replicated five times under identical conditions: the same Kind cluster configuration, the same HPA policy (`stabilizationWindowSeconds: 60` for scale-down), 800 non-reconnecting clients, and the same

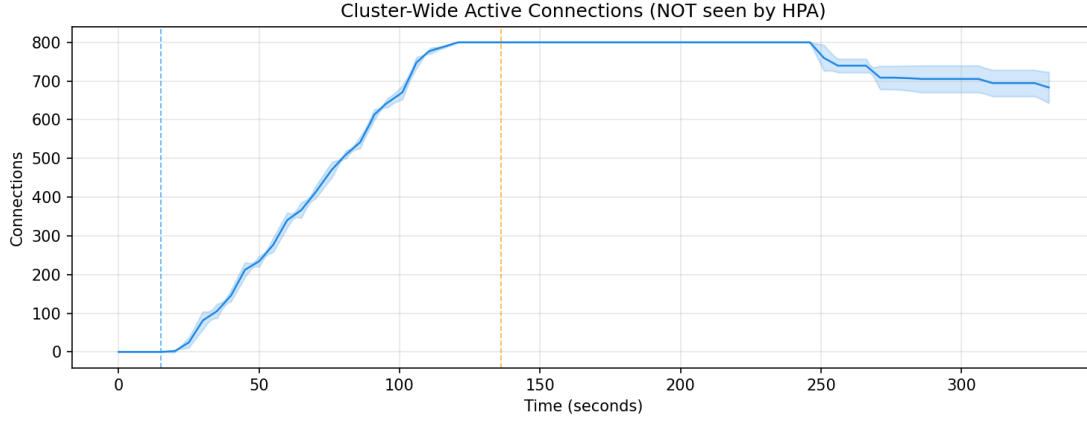


Fig. 7 Experiment B3 (representative Run 4): Active connection count exhibits permanent staircase destruction (800→771→708→644). Each step-down is precisely synchronized with an HPA scale-down event. The connections are irrecoverable under the `reconnect: false` regimen. Across all five replications, the mean final connection count was 649.4 ± 8.8 (Table 1).

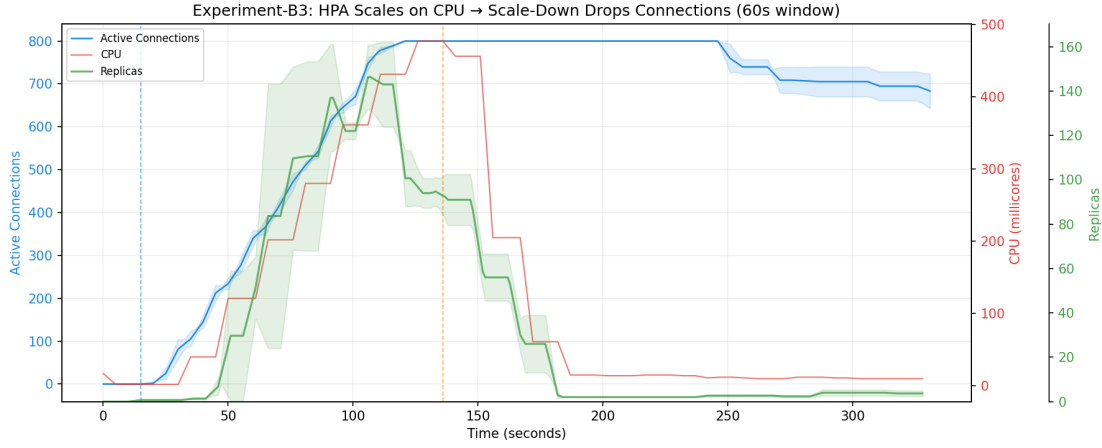


Fig. 8 Experiment B3, combined signal view - the causal proof of the mismatch between CPU-based scaling signals and stateful workload characteristics. CPU utilization (red) drops to near zero as connections become idle → HPA scales down the replica count (green) → active connections (blue) are permanently destroyed. The autoscaler is induced by its own metric to destroy the session state it should be protecting.

two-phase protocol (120-second CONNECT phase followed by a 240-second IDLE phase). Each run used a freshly provisioned cluster to eliminate state carry-over between replications. Table 1 summarizes the per-run connection counts and loss percentages, while Table 2 provides the active connection count at each observed staircase step for every run.

Several observations strengthen the claim that the staircase destruction pattern is a deterministic consequence of HPA’s architecture rather than a stochastic artifact:

1. **Consistency of the destruction magnitude.** The final connection count across all five runs falls within a narrow band of 637–658, yielding a standard deviation of only 8.8 connections (1.1% in loss percentage). This low variance confirms that the number of connections destroyed per scale-down event is governed by the

Table 1 Statistical replication of Experiment B3 (5 runs). Peak connections reached 800 in all runs, while final connection counts exhibited consistent degradation due to scale-down events, averaging 18.8% connection loss per run.

Run	Peak Conn.	Final Conn.	Loss (%)
Run 1	800	637	20.4%
Run 2	800	652	18.5%
Run 3	800	658	17.8%
Run 4	800	644	19.5%
Run 5	800	656	18.0%
Mean ± SD	800 ± 0	649.4 ± 8.8	18.8% ± 1.1%

deterministic connection distribution across pods rather than by random fluctuation.

2. **Qualitative uniformity of the staircase pattern.** Every run exhibits the same three-phase trajectory: (i) monotonic ramp from 0 to 800 during the CONNECT phase, (ii) a

Table 2 Active connection count at each observed staircase step during the IDLE phase for all five B3 replications. Each step-down corresponds to a distinct HPA scale-down event that permanently destroyed the connections hosted on the terminated pod.

Run	Step 0	Step 1	Step 2	Final
Run 1	800	740	–	637
Run 2	800	718	686	652
Run 3	800	761	668	658
Run 4	800	771	708	644
Run 5	800	741	735	656

brief plateau at 800 while CPU is still elevated, and (iii) a staircase descent once CPU drops and HPA begins scaling down. The number of intermediate steps varies between 2 and 3 depending on the timing of HPA’s evaluation cycle relative to the IDLE phase onset, but the qualitative pattern is invariant.

3. **Per-step connection loss.** Each individual scale-down step destroys between 6 and 163 connections, corresponding to the full connection table of the terminated pod. The variation in per-step loss reflects the uneven distribution of connections across pods imposed by Kubernetes’ round-robin Service routing, not any non-determinism in the HPA control loop.

4.5 Summary of the Evidence Chain

Table 3 consolidates the progressive evidence across all four HPA experiments.

5 Discussion

5.1 CPU as an Invalid Scaling Signal for Stateful Workloads

The fundamental architectural mismatch can be stated precisely using two formal definitions. Let $M(t)$ denote the *scaling metric* observed by HPA at time t – in the standard configuration, $M(t)$ is the average CPU utilization across all running replicas. Let $D(t)$ denote the *true workload demand* at time t , defined as the number of active connections $C(t)$ that require server-side session state to be preserved. The HPA proportional control law (Equation 1) computes the desired replica count from $M(t)$, implicitly assuming that the scaling metric is a faithful proxy for workload demand:

$$M(t) \propto D(t) \implies \text{desiredReplicas}(t) \propto D(t) \quad (2)$$

Under this assumption, scaling decisions derived from $M(t)$ correctly track the true capacity requirement. For stateless HTTP workloads where each request consumes a bounded CPU quantum and releases it upon completion, the proportionality $M(t) \propto D(t)$ holds empirically (as confirmed by Experiment A).

For persistent-connection workloads with idle phases, however, this assumption is violated in a specific and destructive manner. During active phases, connections generate CPU work and the proportionality holds; during idle phases, the connections persist (demand $D(t) = C(t) > 0$) but cease generating CPU load, causing the observed metric to collapse:

$$M(t) \rightarrow 0 \quad \text{while} \quad D(t) = C(t) > 0 \quad (3)$$

The consequence is immediate: HPA computes $\text{desiredReplicas} \approx \text{minReplicas}$ (via Equation 1) and initiates scale-down, destroying the connections hosted on the terminated pods. Equation 3 is not a marginal degradation of the proportionality in Equation 2 – it is a complete decoupling of the signal from the demand, producing a scaling decision that is maximally wrong (scale down to minimum while demand is at maximum).

The evidence chain from Experiments B1, B2-Instrumented, and B3 collectively indicates that CPU utilization is a poor proxy for WebSocket workload intensity, and in the cases we tested, an actively misleading one. In Experiment B3, CPU dropped to near-zero precisely because connections were idle, triggering HPA to terminate the pods hosting those connections. This creates a circular destruction pattern: the metric signal drops $\rightarrow$ HPA scales down $\rightarrow$ session state is destroyed. The autoscaler is induced by its own metric to destroy the very resource it should be protecting.

The companion paper demonstrates this converse: a controller operating at CPU_WORK=0 throughout 570 seconds correctly maintains 8 replicas based solely on the connection count signal. This result confirms that the failure resides in the signal, not in the control algorithm: a perfect proportional controller on the wrong signal produces worse outcomes than a simpler controller on the correct signal.

5.2 The Stabilization Window Semantic Mismatch

A natural response to the evidence presented here is to ask whether tuning HPA’s built-in stabilization window to a sufficiently large value would protect idle connections. Our analysis indicates this is unlikely to succeed, for a semantic reason.

HPA’s stabilization window buffers a time series of desiredReplicas values computed from CPU observations. When connections are idle and CPU reads near-zero persistently, every entry in HPA’s window recommends minReplicas . The stabilized output – the maximum over all window entries – therefore converges to minReplicas regardless of how long the window is extended. A longer window merely delays the convergence; it does not retain the information that connections were present.

Table 3 Summary of the progressive evidence chain demonstrating the failure of CPU-based HPA for stateful WebSocket workloads. Each experiment isolates and quantifies a distinct failure mode.

Experiment	Condition	HPA Behavior	Key Quantitative Finding
A (Baseline)	Monotonic load; CPU $\propto$ connections	Correct proportional scaling	Works only under idealized signal correlation
B1 (Cyclic)	Alternating HIGH/LOW activity	Reaches <code>maxReplicas</code> in 99 s	87.5% over-provisioning; 650 s to recover
B2-Inst. (Measured)	Cyclic + Prometheus instrumentation	Reconnection storm per scale-down	Peak 1,400 conn/s; 51.9% connection overshoot
B3 (No-reconnect)	Idle connections; no client retry	Permanent, staircase connection loss	800→649 mean final (18.8% loss); zero recovery

The correct stabilization mechanism must buffer *connection-derived desired counts*: it must retain the high-water mark of recently required capacity, computed from the connection metric rather than from CPU. This is a semantic difference, not a parameter difference. The companion paper implements and evaluates such a mechanism experimentally.

5.3 The 30-Second Limbo Window as a Design Constraint

The Kubernetes pod termination sequence creates a 30-second limbo window between SIGTERM and SIGKILL during which connections survive at the OS level but are doomed. This window is commonly proposed as an opportunity for graceful drain via `preStop` hooks. For WebSocket workloads with session durations of minutes to hours, 30 seconds is typically insufficient for connection migration or drain.

The correct mitigation is not a longer termination period but rather *scale-down suppression* - preventing the scale-down decision from being made in the first place while live sessions are present. Any viable autoscaling solution for persistent-connection workloads must incorporate this principle.

5.4 HPA Stabilization Window Tuning Study

A natural hypothesis is that HPA’s destructive behavior during idle phases could be mitigated by aggressively tuning the `scaleDown.stabilizationWindowSeconds` parameter to outlast the expected idle gap. To evaluate this claim empirically, we conducted a systematic tuning study spanning four stabilization window sizes – 30, 60, 120, and 300 seconds – each tested under two gap conditions.

Experimental Methodology. For each window value $W \in \{30, 60, 120, 300\}$ seconds, we ran the B3 workload protocol (800 non-reconnecting clients, 120-second CONNECT phase with `CPU_WORK=1`) followed by an idle gap of duration G . Two gap conditions were tested per

window: (i) $G < W$ (“below”), where the idle gap is shorter than the stabilization window and connections should be protected; and (ii) $G > W$ (“above”), where the idle gap exceeds the window and HPA is expected to execute a scale-down. The specific gap durations were chosen as $G_{\text{below}} = W - 15$ s and $G_{\text{above}} = W \times 1.5$ to $W + 60$ s, providing margin on both sides of the window boundary. Each of the eight configurations (4 windows $\times$ 2 gap conditions) was replicated three times on freshly provisioned Kind clusters, yielding 24 sub-experiments in total.

Results. Table 4 presents the aggregated results across all configurations. The pattern is unambiguous: when the idle gap falls below the stabilization window, connections are universally preserved (0% loss across all 12 below-window runs). When the gap exceeds the window, connection destruction occurs in every configuration, with mean loss rates ranging from 5.2% (30-second window) to 33.2% (300-second window).

Table 4 HPA stabilization window tuning study results (3 replications per configuration). Gap relation indicates whether the idle gap duration was below or above the configured stabilization window. Connection loss is reported as mean $\pm$ SD across the three runs.

Window	Gap	Relation	Loss (%)	Survived
30s	15s	below	0.0 ± 0.0	3/3
30s	90s	above	5.2 ± 6.2	2/3
60s	45s	below	0.0 ± 0.0	3/3
60s	120s	above	10.2 ± 7.2	2/3
120s	105s	below	0.0 ± 0.3	3/3
120s	180s	above	10.4 ± 5.6	1/3
300s	285s	below	0.0 ± 0.0	3/3
300s	360s	above	33.2 ± 28.0	1/3

Analysis. The results confirm two critical properties of HPA’s stabilization mechanism. First, the stabilization window functions as a *delay*, not a *protection* mechanism: it defers the scale-down decision for W seconds but does not prevent it. As long as CPU remains near zero throughout the

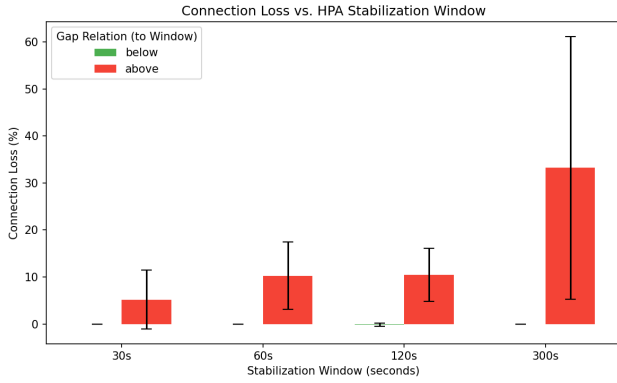


Fig. 9 Connection loss percentage relative to the HPA stabilization window and gap duration. Green bars (“below”) show zero loss when the gap is shorter than the window; red bars (“above”) show consistent connection destruction when the gap exceeds the window, regardless of window size.

window period, every entry in the window’s recommendation buffer converges to `minReplicas`, and the stabilized output (the maximum over the window) eventually equals `minReplicas`.

Second, the increasing variance in the “above” condition at larger window values ($SD = 28.0\%$ at $W = 300s$ vs. $SD = 6.2\%$ at $W = 30s$) reflects the non-deterministic timing of HPA’s 15-second evaluation cycle relative to the window boundary. When the gap only marginally exceeds the window, whether scale-down occurs within the measurement period depends on the alignment of the HPA sync cycle with the gap transition – a source of measurement variance, not of protection.

The tuning study conclusively establishes that the stabilization window is insufficient to protect idle connections: it is a temporal buffer on CPU-derived recommendations, not a connection-aware protection mechanism. No finite window value can prevent scale-down when the underlying metric persistently reads zero.

6 Limitations

1. **Evaluation environment.** All experiments were conducted on a single-machine Kind cluster. While this provides full reproducibility and controlled conditions, it does not capture cross-node scheduling latency, network partitions, or metric-collection noise present in multi-node cloud clusters. Results are directionally valid; absolute timing values (e.g., pod startup lag, HPA sync precision) may differ on production infrastructure.
2. **Workload generalizability.** The load generator uses synthetic WebSocket workloads with a linear connection ramp and deterministic silence gaps. Real-world connection arrival is more bursty and session durations vary widely. For stationary random load (Poisson

arrivals), the proportional failure mechanisms apply unchanged; only the observed peak connection count and storm rate would vary.

3. **Instrumentation dependency.** B2-Instrumented results depend on Prometheus metric freshness at a 15-second scrape interval. A higher scrape interval would affect storm rate measurement precision but would not change the causal finding that scale-down events trigger reconnection cascades.
4. **Single reconnect=false regime.** Experiment B3 uses a uniform `reconnect: false` policy. Real deployments have a mix of reconnecting and non-reconnecting clients; the staircase destruction rate and final session population would differ under a mixed client policy.

7 Conclusion

This paper presented a systematic, empirically grounded characterization of Kubernetes HPA failure modes for stateful WebSocket workloads through a progressive chain of four controlled experiments.

Experiment A established that CPU-based HPA functions correctly under the idealized condition of strict CPU-to-connection proportionality, a scenario rarely achievable in production WebSocket deployments. Experiment B1 demonstrated that cyclic workloads cause HPA to over-provision by **87.5%** (15 vs. the correct 8 replicas), exhausting `maxReplicas` within 99 seconds, with recovery requiring 650 seconds. Experiment B2-Instrumented provided a causal, Prometheus-measured quantification of reconnection storm magnitude, observing a peak rate of **1,400 connections/second** and a 51.9% connection count overshoot directly synchronized with HPA scale-down events. Experiment B3 demonstrated permanent, irrecoverable staircase connection destruction under non-reconnecting clients, with the active session population reduced from **800 to a mean of 649** (18.8% loss, $\pm 1.1\%$ across five replications) with zero recovery, as the combined signal view established the complete causal chain: CPU drop $\rightarrow$ scale-down decision $\rightarrow$ permanent session destruction.

Our experiments suggest that these failure modes are *architectural* rather than configurational: they arise from the limitations of CPU as a scaling signal for persistent-connection workloads, and are unlikely to be remediated by HPA parameter tuning alone. Specifically, the built-in stabilization window does not protect idle connections because it buffers CPU-derived replica recommendations rather than connection-derived desired counts – a semantic mismatch, not a parameter mismatch.

These findings carry a direct design implication: any autoscaling solution for stateful persistent-connection workloads must simultaneously (a) replace the scaling signal with a connection-count metric that encodes actual session

state, and (b) implement scale-down lifecycle semantics that suppress premature termination while live sessions are present. The companion paper presents the design, implementation, and multi-baseline experimental validation of the *StatefulAutoscaler* - a custom Kubernetes controller that addresses all three documented failure modes through connection-density scaling and sliding-window stabilization.

Data Availability

The source code, experimental workloads, Kubernetes manifests, Prometheus configurations, analysis scripts, and processed results required to reproduce Experiments A, B1, B2-Instrumented, and B3 are publicly available at:

`https://github.com/MistaHolmes/ws-k8-controller.git`

The repository includes deployment manifests, load-generation workloads, raw measurement data, figure-generation scripts, and step-by-step replication instructions. All figures presented in this paper were generated directly from the archived experimental results.

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